

MoveIt Random Movement

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1. Environment

Ubuntu 22.04

ROS2: Humble

2. Driving the Real Machine

Driving the real machine involves subscribing to the MoveIt2 `/joint_states` topic to convert the joint state information of the robotic arm into control data for a real robotic arm.

Note: Since the real robotic arm lacks obstacle avoidance, some poses may encounter obstacles; therefore, the planned robotic arm movements should be as reasonable as possible and avoid obstacle-prone areas.

(It is recommended to use preset poses for the demonstration on the real device)

2.1 Starting the Real Device

If not driving the real device, demonstrate the robotic arm movements in MoveIt simulation (skip this startup step):

```
ros2 run dofbot_driver dofbot_driver
```

2.2 Starting MoveIt2

```
ros2 launch dofbot_pro_moveit demo.launch.py
```

3 Random Movement

It is not recommended to drive the real device for the random movement demonstration example, as some poses may encounter obstacles or hit the motherboard! **

Startup Command

After the robotic arm has successfully loaded MoveIt and displays `You can start planning now!`, run the following command: The robotic arm will automatically plan a random motion.

```
ros2 run dofbot_moveit random_move
```

```
[1773991051.493051788]  
4557585124, 68.59035014  
845201443, 179.91510643  
[1773991051.493051788]  
075823116  
5074595, 179.91510643  
[1773991051.493051788]  
1471353516  
1261173, 179.91510643  
[1773991051.493051788]  
029329570  
018238, 179.91510643  
[1773991051.493051788]  
828120273  
42489216, 179.91510643  
[1773991051.493051788]  
043162619  
83926064, 179.91510643  
[1773991051.493051788]  
590981919  
376548, 179.91510643
```

Displays

Robot Alpha	0.5
State Display Time	0.05 s
Use Sim Time	☐
Loop Animation	☐

Planned Path

Add Duplicate Remove Rename

MotionPlanning

ContextPlanningJointsScene ObjectsStored ScenesStored StatesStatus

Commands

PlanExecutePlan & ExecuteStopExecutedClear octomap

Query

Planning Group: grip_group
Start State: <current>
Goal State: close

Options

Planning Time (s): 5.0
Planning Attempts: 10
Velocity Scaling: 0.10
Accel. Scaling: 0.10
☐ Use Cartesian Path
☐ Collision-aware IK
☐ Approx IK Solutions
☐ External Comm.
☐ Replanning
☐ Sensor Positioning

Path Constraints

None

